

Supplementary File for System-Level Co-design of Predictive Controllers and Communication Graphs

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Appendices A, B, C, and D give the observation-model example, path cost proof, joint delivery derivation, and proof of Theorem 2, in the order used by the main paper. Appendices E, F, and G give the settings and verification for the experiments in Sections 5.1, 5.2, and 5.3, respectively.

Appendix A Observation models

This section gives the strict inclusion example cited after Lemma 2 in Section 3.2 of the main paper.

Consider two scalar subsystems with $T = 2$, no communication, no predictions, and

$$A = \begin{bmatrix} 0 & a \\ 0 & 0 \end{bmatrix}, \quad B = I_2, \quad a \neq 0. \quad (\text{A.1})$$

There are no additional response constraints. The state reconstruction masks require both $[\Phi_u]_{0,0}$ and $[\Phi_x]_{1,0}$ to be diagonal. The system-level synthesis (SLS) equations give

$$[\Phi_x]_{1,0} = A + [\Phi_u]_{0,0}, \quad [[\Phi_x]_{1,0}]_{12} = a \neq 0. \quad (\text{A.2})$$

Thus no response satisfies the reconstruction masks. With local disturbance measurements, $\Phi_u = 0$ satisfies every input mask, and

$$\Phi_x = \begin{bmatrix} I_2 & 0 & 0 \\ A & I_2 & 0 \\ A^2 & A & I_2 \end{bmatrix}, \quad \Psi_x = \Psi_u = 0 \quad (\text{A.3})$$

satisfies the SLS equations and temporal constraints. The feasible response set is therefore nonempty under disturbance measurements and empty under the specified reconstruction masks. Both models use the same graph, plant, and response constraints.

Appendix B Path cost recursion

This section proves the recursion for $\kappa_{ij}(\ell)$ in Section 4.2 of the main paper. Assume $c_e > 0$, $d_e \in \{1, 2, \dots\}$, and $\mathcal{Z}_{\text{dep}} = [0, 1]^{|\mathcal{E}_c|}$.

Deleting unused links from a feasible installation leaves a path from j to i without increasing its cost. Thus

$$\kappa_{ij}(\ell) = \min \left\{ \sum_{e \in P} c_e \mid P \text{ is a directed path from } j \text{ to } i, \sum_{e \in P} d_e \leq \ell \right\}. \quad (\text{B.1})$$

The empty path has cost and delay zero, and $\min \emptyset = +\infty$. Positive costs and delays allow all paths in (B.1) to be chosen simple.

We claim that

$$\begin{aligned} \kappa_{ij}(0) &= \begin{cases} 0, & i = j, \\ +\infty, & i \neq j, \end{cases} \\ \kappa_{ij}(\ell) &= \min \left\{ \kappa_{ij}(\ell - 1), \min_{\substack{e=(k,i) \\ d_e \leq \ell}} [\kappa_{kj}(\ell - d_e) + c_e] \right\}, \quad \ell = 1, \dots, T. \end{aligned} \quad (\text{B.2})$$

The initialization follows from $d_e \geq 1$. For the induction step, a path meeting deadline $\ell - 1$ also meets deadline ℓ . Appending $e = (k, i)$ to a path meeting deadline $\ell - d_e$ gives a walk meeting deadline ℓ . Removing any cycle cannot increase cost or delay. Hence $\kappa_{ij}(\ell)$ is no larger than any finite term on the right of (B.2). If $i = j$, both sides are zero. For $i \neq j$, an optimal path either meets deadline $\ell - 1$ or ends in some $e = (k, i)$ with a prefix meeting deadline $\ell - d_e$. Its cost is therefore no smaller than the right-hand side. If no admissible path exists, every term is $+\infty$. This proves (B.2) by induction. □

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Appendix C Joint delivery bounds

This section derives the path allocation dual and proves the joint deployment cost bounds in Section 4.3 of the main paper.

Path formulation. The directed candidate graph is finite, with $c_e > 0$ and positive integer delays d_e . There are no additional deployment constraints. For each m in the finite nonempty set $\mathcal{H}$, let $\mathcal{Q}_m$ contain all simple directed paths from j_m to i_m with total delay at most ℓ_m . Include the empty path when $j_m = i_m$ and $\ell_m \geq 0$, and assume $\mathcal{Q}_m \neq \emptyset$. The minimum cost of meeting all requirements is

$$\begin{aligned} \kappa_{\mathcal{H}} = \min_{z,y} \quad & \sum_{e \in \mathcal{E}_c} c_e z_e \\ \text{subject to} \quad & \sum_{P \in \mathcal{Q}_m} y_{mP} = 1, \quad m \in \mathcal{H}, \\ & \sum_{\substack{P \in \mathcal{Q}_m \\ e \in P}} y_{mP} \leq z_e, \quad m \in \mathcal{H}, e \in \mathcal{E}_c, \\ & z_e, y_{mP} \in \{0, 1\}. \end{aligned} \tag{C.1}$$

Let $a_m = 1$ exactly when the installed graph contains a path in $\mathcal{Q}_m$. A graph meeting every deadline contains a path in $\mathcal{Q}_m$ for each demand. Conversely, the union of one selected path per demand meets every deadline. Removing unused links cannot increase cost. Hence the binary path activation problem and the joint frontier have the same value.

Dual problem. Relax z, y in (C.1) to nonnegative variables and denote the value by $\kappa_{\mathcal{H}}^{\text{LP}}$. Upper bounds on z are unnecessary. Indeed, for any feasible point, replacing z_e by

$$\bar{z}_e = \max_{m \in \mathcal{H}} \sum_{\substack{P \in \mathcal{Q}_m \\ e \in P}} y_{mP} \tag{C.2}$$

preserves feasibility, gives $0 \leq \bar{z}_e \leq 1$, and cannot increase cost. Associate $\lambda_{me} \geq 0$ with the path use constraint for demand m and edge e , and associate a free α_m with its path simplex. The Lagrangian is

$$\begin{aligned} \mathcal{L}(z, y; \alpha, \lambda) = & \sum_m \alpha_m + \sum_e \left(c_e - \sum_m \lambda_{me} \right) z_e \\ & + \sum_m \sum_{P \in \mathcal{Q}_m} \left(\sum_{e \in P} \lambda_{me} - \alpha_m \right) y_{mP}. \end{aligned} \tag{C.3}$$

Its infimum over nonnegative z, y is finite exactly when

$$\sum_m \lambda_{me} \leq c_e, \quad \alpha_m \leq \sum_{e \in P} \lambda_{me} \quad \text{for every } m, P \in \mathcal{Q}_m. \tag{C.4}$$

This is the allocation dual in the main paper. The finite path LP is feasible and its objective is bounded below, so finite dimensional linear programming duality applies. Every path charge is nonnegative. Therefore, replacing any negative α_m by zero preserves feasibility and cannot decrease the objective, which proves that $\alpha_m \geq 0$ is lossless.

Availability inequalities. Fix any dual-feasible allocation and an installed graph. Let $\mathcal{A} = \{m \in \mathcal{H} \mid a_m = 1\}$. If $i_m = j_m$, choose the zero-length path, which belongs to $\mathcal{Q}_m$. Otherwise, exact reachability supplies an installed deadline-feasible walk, and positive link latencies permit cycle removal without increasing its latency. Hence an installed simple path $P_m \in \mathcal{Q}_m$ exists for every $m \in \mathcal{A}$. Thus

$$\begin{aligned} \sum_m \alpha_m a_m & \leq \sum_{m \in \mathcal{A}} \sum_{e \in P_m} \lambda_{me} = \sum_e \sum_{\substack{m \in \mathcal{A} \\ e \in P_m}} \lambda_{me} \\ & \leq \sum_e z_e \sum_m \lambda_{me} \leq \sum_e c_e z_e. \end{aligned} \tag{C.5}$$

The third relation uses binary installation and $\lambda \geq 0$. All allocations on one edge sum to at most its cost, including when requirements share that edge. The budget gives $\sum_m \alpha_m a_m \leq C$. If $\sum_m \alpha_m > C$, not all binary indicators can equal one. Thus

$$\sum_{m \in \mathcal{H}} a_m \leq |\mathcal{H}| - 1. \tag{C.6}$$

If the LP value exceeds C , dual attainment supplies such an allocation. A numerical allocation gives a valid inequality only after every edge and path constraint has been verified.

Deployment cost bounds. Here $\kappa_{i_m j_m}(\ell_m)$ is the minimum cost of a path from j_m to i_m with delay at most ℓ_m . Fix a relaxed primal point and a requirement m . Set $u_{me} = \sum_{P \ni e} y_{mP}$. The path simplex and $z_e \geq u_{me}$ give

$$\sum_e c_e z_e \geq \sum_e c_e u_{me} = \sum_{P \in \mathcal{Q}_m} y_{mP} \sum_{e \in P} c_e$$

$$\geq \min_{P \in \mathcal{Q}_m} \sum_{e \in P} c_e = \kappa_{i_m j_m}(\ell_m). \quad (\text{C.7})$$

The last equality uses pure deployment and positive costs. Taking the maximum over demands and then the LP infimum proves the first lower bound. Relaxing the binary path problem proves the second, while installing any chosen path union proves the upper bound. For a supplied dual feasible allocation and path collection, weak duality, relaxation, and path union feasibility give

$$\sum_m \alpha_m \leq \kappa_{\mathcal{H}}^{\text{LP}} \leq \kappa_{\mathcal{H}} \leq \sum_{e \in \cup_m P_m} c_e. \quad (\text{C.8})$$

Equality of the endpoint values forces equality in the last three relations. It certifies relaxation exactness and joint optimality of the path union, but not graph uniqueness.

Appendix D Proof of the control performance theorem

This section proves Theorem 2 of the main paper.

Proof. Throughout, the co-design problem is feasible, $\mathcal{C}_{\Theta}$ is closed and convex, $Q \succeq 0$, $R \succ 0$, and $\Sigma \succ 0$. Only the budget restricts link installation, and relaying is allowed. The observation model, release times, joint second moment Σ , and response bounds are fixed.

Full graph optimum. Recall that $\mathcal{F}(z)$ is the feasible response set for graph z and $a_g(z)$ is the availability indicator of group g . Write $\Theta_x = [\Phi_x \ \Psi_x]$ and $\Theta_u = [\Phi_u \ \Psi_u]$ as in the main paper. Since

$$\mathcal{F}(z) \subseteq \mathcal{F}(\mathbf{1}), \quad z \in \mathcal{Z}_C,$$

the full graph problem is feasible. The SLS equations and control objective give

$$\Theta_x = [\mathcal{D}^{-1} \ 0] + \mathcal{D}^{-1} \mathcal{B} \Theta_u, \quad (\text{D.1})$$

$$\begin{aligned} J_{\text{sls}}(\Theta; \Sigma) &= \text{tr} \left[\left(\Theta_x^T Q \Theta_x + \Theta_u^T R \Theta_u \right) \Sigma \right] \\ &\geq \lambda_{\min}(\Sigma) \lambda_{\min}(R) \|\Theta_u\|_F^2. \end{aligned} \quad (\text{D.2})$$

After eliminating Θ_x , the feasible set is closed and convex. The objective is coercive and strictly convex in Θ_u . Thus the full graph has a unique optimal response Θ^{all} , with cost $V(\mathbf{1})$. Write g^{all} for its group g .

Lower bound. Fix $z \in \mathcal{Z}_C$ and $\Theta \in \mathcal{F}(z)$. Set $\Delta\Theta_x = \Theta_x - \Theta_x^{\text{all}}$ and $\Delta\Theta_u = \Theta_u - \Theta_u^{\text{all}}$. Optimality on the convex set $\mathcal{F}(\mathbf{1})$ gives

$$\text{tr} \left[\left((\Theta_x^{\text{all}})^T Q \Delta\Theta_x + (\Theta_u^{\text{all}})^T R \Delta\Theta_u \right) \Sigma \right] \geq 0. \quad (\text{D.3})$$

The quadratic expansion is

$$\begin{aligned} J_{\text{sls}}(\Theta; \Sigma) - V(\mathbf{1}) &= 2 \text{tr} \left[\left((\Theta_x^{\text{all}})^T Q \Delta\Theta_x + (\Theta_u^{\text{all}})^T R \Delta\Theta_u \right) \Sigma \right] + \text{tr} \left[\left(\Delta\Theta_x^T Q \Delta\Theta_x + \Delta\Theta_u^T R \Delta\Theta_u \right) \Sigma \right] \\ &\geq \lambda_{\min}(\Sigma) \lambda_{\min}(R) \|\Delta\Theta_u\|_F^2, \end{aligned} \quad (\text{D.4})$$

where the last step uses (D.3). The first-order term need not be zero. Equation (D.1) gives $\Delta\Theta_x = \mathcal{D}^{-1} \mathcal{B} \Delta\Theta_u$. An unavailable group satisfies $g = 0$. Since the response groups are disjoint,

$$\begin{aligned} \sum_{g \in \mathcal{I}} (1 - a_g(z)) \|g^{\text{all}}\|_2^2 &= \sum_{g \in \mathcal{I}} (1 - a_g(z)) \|g - g^{\text{all}}\|_2^2 \\ &\leq \|\Delta\Theta_x\|_F^2 + \|\Delta\Theta_u\|_F^2 \\ &\leq (1 + \|\mathcal{D}^{-1} \mathcal{B}\|_2^2) \|\Delta\Theta_u\|_F^2. \end{aligned} \quad (\text{D.5})$$

In the reconstruction model, the identity terms in $\Phi_x - I$ cancel in the difference. The sum includes the terminal state groups. Combining (D.4) and (D.5), then taking the infimum over $\Theta \in \mathcal{F}(z)$ and the minimum over $z \in \mathcal{Z}_C$, gives

$$J^* - V(\mathbf{1}) \geq \frac{\lambda_{\min}(\Sigma) \lambda_{\min}(R)}{1 + \|\mathcal{D}^{-1} \mathcal{B}\|_2^2} \min_{z \in \mathcal{Z}_C} \sum_{g \in \mathcal{I}} (1 - a_g(z)) \|g^{\text{all}}\|_2^2. \quad (\text{D.6})$$

Upper bound. Now assume direct disturbance observation, no additional constraints in $\mathcal{C}_{\Theta}$, and $M_g > 0$ for every $g \in \mathcal{I}$. In the full graph problem, write the group constraints as

$$\|g\|_2^2 \leq M_g^2, \quad g = 0 \quad \text{if } a_g(\mathbf{1}) = 0. \quad (\text{D.7})$$

The response $\Theta_u = 0$, $\Theta_x = [\mathcal{D}^{-1} \ 0]$ satisfies all affine constraints and obeys $\|g\|_2^2 = 0 < M_g^2$ for every $g \in \mathcal{I}$. Thus Slater's condition holds on the affine feasible set. Let $\nu_g \geq 0$ be the optimal KKT multipliers of the squared ball constraints in (D.7). Complementary slackness gives

$$\nu_g (\|g^{\text{all}}\|_2^2 - M_g^2) = 0. \quad (\text{D.8})$$

Fix any $z \in \mathcal{Z}_C$. Construct $\hat{\Theta}_u$ from Θ_u^{all} by setting

$$\hat{g} = a_g(z)g^{\text{all}}, \quad g \in \mathcal{I}, \quad (\text{D.9})$$

and keeping all input coefficients outside these groups unchanged. Set

$$\hat{\Theta}_x = [\mathcal{D}^{-1} \ 0] + \mathcal{D}^{-1} \mathcal{B} \hat{\Theta}_u. \quad (\text{D.10})$$

This construction preserves the input masks, temporal zeros, and group bounds. Equation (D.10) enforces the SLS equations and the state temporal masks. No communication masks are imposed on the state responses. Hence $\hat{\Theta} \in \mathcal{F}(z)$. Take $\Theta = \hat{\Theta}$ in the definitions of $\Delta\Theta_x$ and $\Delta\Theta_u$. The groups are disjoint and the indicators are binary, giving

$$\|\Delta\Theta_u\|_F^2 = \sum_{g \in \mathcal{I}} (1 - a_g(z)) \|g^{\text{all}}\|_2^2. \quad (\text{D.11})$$

The difference satisfies the homogeneous linear equations of the full graph problem. The equality multiplier terms therefore vanish when stationarity is evaluated along this difference. The KKT conditions give

$$\begin{aligned} 2 \operatorname{tr} \left[\left((\Theta_x^{\text{all}})^\top Q \Delta\Theta_x + (\Theta_u^{\text{all}})^\top R \Delta\Theta_u \right) \Sigma \right] &= -2 \sum_{g \in \mathcal{I}} \nu_g \langle g^{\text{all}}, \hat{g} - g^{\text{all}} \rangle \\ &= 2 \sum_{g \in \mathcal{I}} \nu_g (1 - a_g(z)) \|g^{\text{all}}\|_2^2. \end{aligned} \quad (\text{D.12})$$

Substituting this identity and $\Delta\Theta_x = \mathcal{D}^{-1} \mathcal{B} \Delta\Theta_u$ into the quadratic expansion yields

$$\begin{aligned} J_{\text{sls}}(\hat{\Theta}; \Sigma) - V(\mathbf{1}) &= 2 \sum_{g \in \mathcal{I}} \nu_g (1 - a_g(z)) \|g^{\text{all}}\|_2^2 + \operatorname{tr} \left[\Delta\Theta_u^\top \left(R + \mathcal{B}^\top \mathcal{D}^{-\top} Q \mathcal{D}^{-1} \mathcal{B} \right) \Delta\Theta_u \Sigma \right] \\ &\leq 2 \sum_{g \in \mathcal{I}} \nu_g (1 - a_g(z)) \|g^{\text{all}}\|_2^2 + \lambda_{\max}(\Sigma) \lambda_{\max} \left(R + \mathcal{B}^\top \mathcal{D}^{-\top} Q \mathcal{D}^{-1} \mathcal{B} \right) \|\Delta\Theta_u\|_F^2. \end{aligned} \quad (\text{D.13})$$

The last inequality bounds the trace using the largest eigenvalues. It retains the full joint second moment Σ . The terms $2\nu_g$ account for active group bounds. For each $z \in \mathcal{Z}_C$, feasibility gives $J^* \leq V(z) \leq J_{\text{sls}}(\hat{\Theta}; \Sigma)$. Use (D.11) and minimize the last line of (D.13) over z . Together with the direct observation lower bound, this proves

$$\begin{aligned} \lambda_{\min}(\Sigma) \lambda_{\min}(R) \min_{z \in \mathcal{Z}_C} \sum_{g \in \mathcal{I}} (1 - a_g(z)) \|g^{\text{all}}\|_2^2 &\leq J^* - V(\mathbf{1}) \\ &\leq \min_{z \in \mathcal{Z}_C} \sum_{g \in \mathcal{I}} (1 - a_g(z)) \left[\lambda_{\max}(\Sigma) \lambda_{\max} \left(R + \mathcal{B}^\top \mathcal{D}^{-\top} Q \mathcal{D}^{-1} \mathcal{B} \right) + 2\nu_g \right] \|g^{\text{all}}\|_2^2. \end{aligned} \quad (\text{D.14})$$

A graph attaining the upper minimum, with the response in (D.9) and (D.10), gives a feasible controller satisfying that bound.

Budget threshold. Return to the initial assumptions, without the upper-bound restrictions. The set $\mathcal{H}$ consists exactly of the delivery requirements of groups with $g^{\text{all}} \neq 0$, with $\kappa_\emptyset = 0$. By the definition of $\kappa_{\mathcal{H}}$,

$$\begin{aligned} C \geq \kappa_{\mathcal{H}} &\iff \exists z \in \mathcal{Z}_C : a_g(z) = 1 \text{ for every } g \in \mathcal{I} \text{ with } g^{\text{all}} \neq 0 \\ &\iff \Theta^{\text{all}} \in \bigcup_{z \in \mathcal{Z}_C} \mathcal{F}(z). \end{aligned} \quad (\text{D.15})$$

The last equivalence holds because all other response constraints are fixed. Thus $C \geq \kappa_{\mathcal{H}}$ gives $J^* = V(\mathbf{1})$. If $C < \kappa_{\mathcal{H}}$, every graph in $\mathcal{Z}_C$ makes at least one nonzero group of Θ^{all} unavailable. Since $\mathcal{Z}_C$ is finite and nonempty,

$$\min_{z \in \mathcal{Z}_C} \sum_{g \in \mathcal{I}} (1 - a_g(z)) \|g^{\text{all}}\|_2^2 > 0.$$

Equation (D.6) then gives $J^* > V(\mathbf{1})$. We have proved

$$J^* = V(\mathbf{1}) \iff C \geq \kappa_{\mathcal{H}}. \quad (\text{D.16})$$

which completes the proof $\square$

Appendix E Shared route experiment

For the experiment in Section 5.1, fixed-graph synthesis and the reported lower bounds are evaluated in rational arithmetic.

Controller synthesis. Set $a = 4/5$, $b = 2/5$, and $u_{i,t} = g_{i,t}\xi$ for $i \in \{3, 4\}$. For a graph G , let $D_G(1, i)$ be its shortest path latency, with value $+\infty$ if no path exists, and let $I = \{k \in \{0, 1, 2, 3\} : k \geq D_G(1, i)\}$. Writing $g = (g_{i,0}, \dots, g_{i,3})^\top$ gives $x_{i,t} = (\beta_t + d_t^\top g)\xi$, where

$$\beta_0 = 0, \quad \beta_t = bta^{t-1} \ (t \geq 1), \quad [d_t]_k = \begin{cases} a^{t-1-k}, & k < t, \\ 0, & k \geq t. \end{cases}$$

The receiver cost is

$$\begin{aligned} f_I(g_I) &= g_I^\top H_I g_I + 2v_I^\top g_I + \sum_{t=0}^4 \beta_t^2, \\ H_I &= \sum_{t=0}^4 d_{t,I} d_{t,I}^\top + \frac{1}{10} I_{|I|}, \quad v_I = \sum_{t=0}^4 \beta_t d_{t,I}. \end{aligned} \quad (\text{E.1})$$

For nonempty I , $H_I \succ 0$ gives $g_I = -H_I^{-1} v_I$. For $I = \emptyset$, the input is zero. All 16 affordable graphs satisfy $|g_{i,t}| \leq 2$ at these minimizers, so the unconstrained and bounded optima coincide. The two receiver costs add. For the optimal shared route,

$$J^* = \frac{1277832}{1015625}, \quad g_{i,2} = -\frac{8128}{11375}, \quad g_{i,3} = -\frac{512}{2275}.$$

Greedy refitting resolves equal cost improvements by smaller deployment cost, then by the edge order $1 \rightarrow 2, 2 \rightarrow 3, 2 \rightarrow 4, 1 \rightarrow 3, 1 \rightarrow 4$. Local search tests all affordable single-link additions, deletions, and exchanges.

Relaxations. Let v_i denote availability through the relay and a_i availability by time two. For $i \in \{3, 4\}$, impose

$$z_{1 \rightarrow 2} + z_{2 \rightarrow 3} + z_{2 \rightarrow 4} + 2z_{1 \rightarrow 3} + 2z_{1 \rightarrow 4} \leq 3, \quad (\text{E.2})$$

$$v_i \leq z_{1 \rightarrow 2}, \quad v_i \leq z_{2 \rightarrow i}, \quad v_i \geq z_{1 \rightarrow 2} + z_{2 \rightarrow i} - 1, \quad (\text{E.3})$$

$$a_i \geq z_{1 \rightarrow i}, \quad a_i \geq v_i, \quad a_i \leq z_{1 \rightarrow i} + v_i, \quad (\text{E.4})$$

with all installation and availability variables in $[0, 1]$. Set $a_{i,1} = z_{1 \rightarrow i}$ and $a_{i,2} = a_{i,3} = a_i$. All relaxations retain $g_{i,0} = 0$ and $|g_{i,t}| \leq 2a_{i,t}$. The continuous relaxation uses the original quadratic cost. The perspective relaxation replaces each input penalty $g_{i,t}^2/10$ by $g_{i,t}^2/(10a_{i,t})$, with value zero at $(0, 0)$. The joint cover relaxation also imposes

$$a_{3,1} + a_{4,1} \leq 1, \quad a_{3,1} + a_{4,2} \leq 1, \quad a_{4,1} + a_{3,2} \leq 1.$$

Each excluded pair requires cost at least four. Enumeration verifies these inequalities at budget three.

Lower bounds. Collect the relaxation variables in w , with rational constraints $Aw \leq d$ and $l \leq w \leq h$. Gain bounds are $[-2, 2]$ and installation and availability bounds are $[0, 1]$. Quadratic tangents and

$$\frac{g^2}{10a} \geq \frac{\bar{g}}{5\bar{a}}g - \frac{\bar{g}^2}{10\bar{a}^2}a, \quad \bar{a} > 0,$$

give an affine objective minorant $c_0 + c^\top w$. Tangent points are obtained from numerical solutions and converted to rationals, with $\bar{a} \geq 10^{-6}$. For rational $\lambda \geq 0$ and every feasible w ,

$$L = c_0 - \lambda^\top d + \sum_j \min\{(c + A^\top \lambda)_j l_j, (c + A^\top \lambda)_j h_j\} \leq c_0 + c^\top w. \quad (\text{E.5})$$

The inequality follows by adding $\lambda^\top (Aw - d) \leq 0$ and minimizing over the box. Numerical multipliers are converted to rationals and clipped at zero. The box minimum accounts for stationarity errors. Reported lower bounds are rounded downward from L .

Predictions. The disturbance occurs at absolute time two, with zero states and inputs beforehand. Write $\xi = p + \epsilon$ with independent p and ϵ satisfying

$$\mathbb{E}[p] = \mathbb{E}[\epsilon] = 0, \quad \mathbb{E}[p^2] = 3/4, \quad \mathbb{E}[\epsilon^2] = 1/4.$$

The source observes p at time $2 - q$ and ξ at time two, where $q \in \{0, 1, 2\}$. Relative to the disturbance event, the first admissible input indices are $D_G(1, i)$ for ϵ and $\max\{0, D_G(1, i) - q\}$ for p . When ξ arrives, the receiver already has p and forms $\epsilon = \xi - p$. Applying the same quadratic minimization to the two independent signals gives

$$u = g^\epsilon \epsilon + g^p p = g^\epsilon \xi + (g^p - g^\epsilon)p, \quad \Phi_u = g^\epsilon, \quad \Psi_u = g^p - g^\epsilon.$$

The expected costs have weights $1/4$ and $3/4$. Every scalar entry of Φ_u and Ψ_u satisfies the bound two for all affordable graphs and all three release rules.

Appendix F Performance bound experiment

Parameters. The experiment in Section 5.2 uses $T = 4$ and

$$A_t = \begin{bmatrix} 0.70 & 0.06 & 0.04 \\ 0.26 & 0.64 & 0.08 \\ 0.17 & 0.23 & 0.66 \end{bmatrix}, \quad B_t = \text{diag}(0.50, 0.55, 0.60). \quad (\text{F.1})$$

The stage state weight is $\text{diag}(1, 1.2, 1.5)$, the terminal weight is twice this matrix, and the input weight is $0.2I_3$. Disturbances and predictions are jointly Gaussian with zero mean and

$$\Sigma = \begin{bmatrix} I_{15} & 0.8I_{15} \\ 0.8I_{15} & I_{15} \end{bmatrix}.$$

The four settings combine $\rho_s = \max\{0, s - q\}$, $q \in \{0, 1\}$, with uniform scalar input response bounds $M_g = 2$ or $M_g = 0.1$. Disturbances are measured locally and relaying is allowed. There are no state communication masks or additional response constraints.

Evaluation. For each setting, all 32 graphs are evaluated by convex synthesis. Clarabel uses feasibility, absolute gap, and relative gap tolerances 10^{-10} . Each cost or bound is minimized separately over $\mathcal{Z}_C$, using the full-graph optimum for the same release rule and M_g . The constructed controller uses (D.9) and (D.10). The analytical bounds are (D.14), with lower-bound coefficient $\lambda_{\min}(\Sigma)\lambda_{\min}(R) = 0.04$. For the normalized constraint $(g/M_g)^2 \leq 1$, dividing its multiplier by M_g^2 gives ν_g for $g^2 \leq M_g^2$.

Numerical accuracy. SLS residuals, input masks, response bounds, and bound ordering are checked within 2×10^{-6} . Over available input coefficients, the maximum absolute stationarity residual is divided by one plus the maximum absolute objective gradient. This normalized residual and absolute complementarity errors are at most 10^{-5} , and $\nu_g \geq -10^{-5}$. The threshold $\kappa_{\mathcal{H}} = 5$ is unchanged when the support tolerance varies over $10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}$. These are floating-point checks, not rational certificates.

Appendix G Voltage experiments

This section gives the data, synthesis constraints, and cost certificates for Section 5.3 of the main paper.

Data. The voltage model uses

$$A_v = 0.80I + 0.035D^{-1/2}A_fD^{-1/2}, \quad B_v = -0.31 \frac{X_{\text{net}}}{\|X_{\text{net}}\|_2}, \quad (\text{G.1})$$

where A_f is the adjacency matrix of the controlled buses and D is its diagonal degree matrix. The entry $[X_{\text{net}}]_{ij}$ is twice the total reactance shared by the paths from the slack bus to buses i and j . Table G1 specifies the eight-bus feeder, with parent zero denoting the slack bus. Disturbances are centered and independent across buses and times. The two cases differ only in their standard deviations σ_j . Both use $T = 5$, $Q = 22I$, $R = 0.35I$, and $\Psi_x = \Psi_u = 0$.

Table G1 Eight bus feeder and disturbance data.

Bus	Parent	Reactance	Baseline deviation	Residual deviation
1	0	0.020	0.017	0.009
2	1	0.024	0.020	0.011
3	1	0.018	0.016	0.010
4	2	0.030	0.024	0.013
5	2	0.022	0.019	0.012
6	3	0.027	0.022	0.010
7	3	0.025	0.018	0.011
8	6	0.031	0.026	0.014

Synthesis. With local disturbance measurements and direct transmission, write $X_{t,s} = [\Phi_x]_{t,s}$ and $U_{t,s} = [\Phi_u]_{t,s}$, with zero blocks for $t < s$. The plant equations and one-period link latency require

$$X_{s,s} = I, \quad X_{t+1,s} = A_v X_{t,s} + B_v U_{t,s}, \quad [U_{t,t}]_{ij} = 0 \quad (i \neq j), \quad (\text{G.2})$$

for $0 \leq s \leq t < T$, with $X_{T,T} = I$. For each remote link $j \rightarrow i$, impose

$$g_{ij} = \text{col}\{[U_{t,s}]_{ij} \mid 0 \leq s < t < T\}, \quad \|g_{ij}\|_2 \leq 5z_{ij}. \quad (\text{G.3})$$

Set $z_{ij} = 0$ for noncandidate links. Local coefficients are unbounded, and X has no spatial masks. There are no voltage, inverter, or ramp limits. The objective is

$$22 \sum_{0 \leq s \leq t \leq T} \|X_{t,s} \text{diag}(\sigma)\|_F^2 + 0.35 \sum_{0 \leq s \leq t < T} \|U_{t,s} \text{diag}(\sigma)\|_F^2. \quad (\text{G.4})$$

The perspective relaxation replaces each remote input penalty by θ_{ij} and imposes

$$0.35\sigma_j^2 \|g_{ij}\|_2^2 \leq \theta_{ij} z_{ij}, \quad \theta_{ij} \geq 0, \quad 0 \leq z_{ij} \leq 1. \quad (\text{G.5})$$

It retains (G.2), (G.3), $\sum_{i \neq j} z_{ij} \leq C$, and the state and local input costs. ECOS uses feasibility and gap tolerances 10^{-9} for the relaxation. Clarabel uses feasibility and absolute gap tolerances 10^{-7} for fixed-graph synthesis.

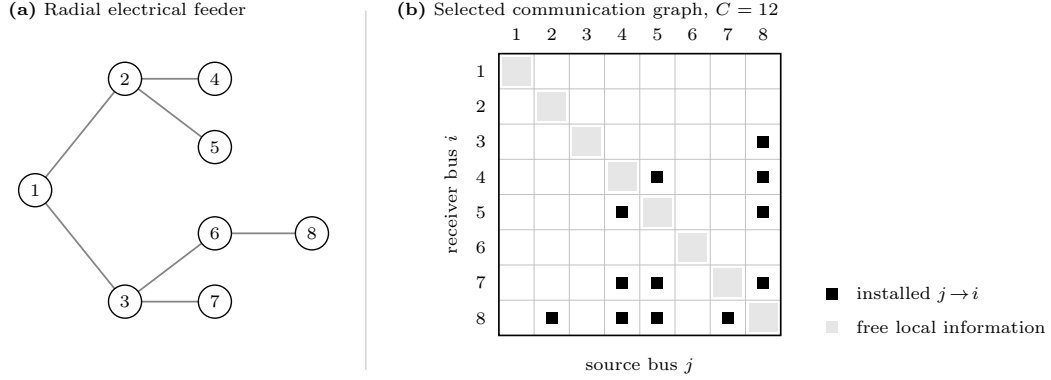


Figure G1 Electrical feeder and communication graph selected by Perspective at $C = 12$ under the residual covariance. (a) The eight controlled buses and feeder branches. (b) The communication matrix $Z = [z_{ij}]$. A black square at row i and column j denotes the installed link $j \rightarrow i$. The gray diagonal denotes free local information.

Graph selection. Perspective ranks the relaxed z_{ij} . Magnitude ranks $\|g_{ij}^{\text{all}}\|_2$ from the bounded full-graph optimum. Sensitivity ranks $\|[X_{\text{net}}]_{ij}\|/\|X_{\text{net}}\|_2$. Distance ranks the negative feeder branch count plus $0.02\|[X_{\text{net}}]_{ij}\|/\|X_{\text{net}}\|_2$. Each method selects the highest scores and refits the controller. Greedy refits every candidate single-link addition. It treats costs within 10^{-6} times the largest of one, the current cost, and the lowest candidate cost as ties. All remaining ties use receiver index, then source index.

Additional results. Table G2 reports cost reductions under both covariances and the overlap between the Perspective graphs. The graphs coincide at $C = 4$ and differ at the other four budgets. Figure G1 shows the residual-covariance graph at $C = 12$. None of its selected links joins electrically adjacent buses. Bus 8 receives from 2, 4, 5, 7 and sends to 3, 4, 5, 7. These are properties of the selected graph, not optimality claims.

Table G2 Ranking results under the two prescribed disturbance covariances. The entries for P, M, S, and D are $100(J_{\text{local}} - J(C))/(J_{\text{local}} - J_{\text{full}})$, where $J(C)$ is the refitted cost at budget C . The zero-link and full-graph references use the same covariance. P denotes Perspective, M Magnitude, S Sensitivity, and D Distance. O is the number of links common to the two Perspective graphs divided by the number of links in their union.

Budget	Baseline covariance				Residual covariance				O
	P	M	S	D	P	M	S	D	
4	47.8	40.0	10.9	10.9	49.5	41.1	10.5	10.5	1.000
8	65.9	46.8	20.3	15.1	65.7	47.9	20.3	14.7	0.778
12	77.8	60.2	33.2	16.4	75.0	59.9	32.6	16.1	0.846
16	83.1	69.4	49.3	25.1	85.1	71.2	48.5	25.3	0.684
24	92.6	86.6	59.6	49.3	91.6	86.8	59.7	49.3	0.846

Table G3 compares the relaxation's dual estimate with the refitted controller cost. The refitted cost minus the dual estimate, divided by the refitted cost, ranges from 0.024% to 0.136%. These floating-point differences are not verified optimality gaps. The certificates below instead recompute bounds in rational arithmetic.

Table G3 Floating-point solver outputs under the residual covariance. Gap reduction uses the normalization in Table G2. The final column reports the relative difference defined in the text, not a verified optimality gap.

Links	ECOS dual estimate	Fixed graph cost	Gap reduction percent	Relative difference percent
4	0.232641	0.232958	49.5	0.136
8	0.231983	0.232282	65.7	0.129
12	0.231638	0.231896	75.0	0.111
16	0.231418	0.231476	85.1	0.025
24	0.231148	0.231204	91.6	0.024

Verification. For both voltage certificates, the generated binary64 entries of A_v , B_v , and σ are treated as exact rationals, with variances σ_j^2 and input weight 35/100. Unavailable input coefficients are zeroed. State responses, costs, gradients, and norms are then recomputed exactly using (G.2). An upper bound requires $\|g_{ij}\|_2^2 \leq 25$ for every remote group. All reported intervals use directed rounding.

Five-bus certificate. The line feeder has branch reactances (0.018, 0.0215, 0.025, 0.0285, 0.032) from the slack bus and disturbance standard deviations (0.014, 0.017, 0.015, 0.020, 0.018). Use (G.1), $T = 3$, $Q = 22I$, $R = 0.35I$, and $C = 3$. The eight unit-cost candidate

links join adjacent controlled buses in both directions. Let $\mathcal{H}_3$ contain the 93 affordable graphs. For $G \in \mathcal{H}_3$, eliminate X by (G.2) and let w contain the input coefficients allowed by G . Without group bounds, the cost is

$$f_G(w) = 22 \sum_{0 \leq s \leq t \leq T} \|X_{t,s}(w) \text{diag}(\sigma)\|_F^2 + 0.35 \sum_{0 \leq s \leq t < T} \|U_{t,s}(w) \text{diag}(\sigma)\|_F^2. \quad (\text{G.6})$$

Let V_G be the bounded optimum and $J_5^* = \min_{G \in \mathcal{H}_3} V_G$. Since $\nabla^2 f_G \succeq 2(0.35)\sigma_{\min}^2 I$, where $\sigma_{\min} = \min_j \sigma_j > 0$, any $\bar{w}_G$ gives

$$\ell_G := f_G(\bar{w}_G) - \frac{\|\nabla f_G(\bar{w}_G)\|_2^2}{4(0.35)\sigma_{\min}^2} \leq \min_w f_G(w) \leq V_G. \quad (\text{G.7})$$

Thus a feasible graph $\hat{G}$ with verified cost $\bar{J}_5$ satisfies

$$\min_{G \in \mathcal{H}_3} \ell_G \leq J_5^* \leq \bar{J}_5, \quad \underline{J}_e^b = \min_{\substack{G \in \mathcal{H}_3 \\ \mathbf{1}_{\{e \in G\}} = b}} \ell_G, \quad b \in \{0, 1\}. \quad (\text{G.8})$$

A link decision b is excluded whenever $\underline{J}_e^b > \bar{J}_5$. All 93 reconstructed responses satisfy the group bounds. The verified inequalities are

$$0 \leq \bar{J}_5 - J_5^* \leq 2.6 \times 10^{-32}, \quad \underline{J}_e^{1-\mathbf{1}_{\{e \in \hat{G}\}}} - \bar{J}_5 \geq 1.07 \times 10^{-5} > 0 \quad \text{for every candidate } e. \quad (\text{G.9})$$

Every graph other than $\hat{G}$ differs in an excluded link decision, so $\hat{G}$ is uniquely optimal.

Eight-bus certificate. Use the eight-bus model with $T = 5$, the residual deviations in Table G1, and $C = 12$. Let f_{f} be (G.6) with all delayed remote coefficients free and the same-time remote zeros retained. Dropping the budget and group bounds gives

$$\begin{aligned} \underline{J}_{\text{f}} &:= f_{\text{f}}(\bar{w}) - \frac{\|\nabla f_{\text{f}}(\bar{w})\|_2^2}{4(0.35) \min_j \sigma_j^2} \leq \min_w f_{\text{f}}(w) \leq J_{8,12}^* \\ &\leq f_{\text{f}}(\hat{w}) = \bar{J}_{12}, \end{aligned} \quad (\text{G.10})$$

where $\hat{w}$ is the verified controller on the twelve-link Perspective graph, with unavailable coordinates set to zero. The residual correction in the lower bound is at most 7.3×10^{-32} . The resulting interval is

$$0.2308569050885360 \leq J_{8,12}^* \leq 0.2318963870690745, \quad 0 \leq \frac{\bar{J}_{12} - J_{8,12}^*}{\bar{J}_{12}} \leq 0.449\%. \quad (\text{G.11})$$

Its width is at most 1.04×10^{-3} . This bounds controller suboptimality without certifying the selected graph as optimal.